

1. PROJECT PROFILE

PROJECT TITLE	Supermart Grocery Sales – Retail Analytics
OBJECTIVE	• Identify key factors driving sales. • Optimize inventory management. • Improve customer satisfaction. • Discover frequent purchase patterns. • Forecast future sales trends to support planning and growth strategies. • Observe Sales Patterns over time and Regional Trends. • Effective business decision-making.
PLATFORM	Windows
TOOLS AND TECHNOLOGIES	Excel (Version – 19H2) Python (Version –3.13) Jupyter (Version–7.3.2)
PROJECT DURATION	20 Days
DEVELOPED BY	Ms. Varsha

2. INTRODUCTION

- This project analyzes a fictional dataset of grocery delivery orders in Tamil Nadu, India. The dataset provides insights into customer purchasing behavior, sales patterns and regional trends.
- Using machine learning algorithms this project identifies key factors influencing sales, optimize inventory management, and enhance customer satisfaction.
- Additionally, by employing analysis and visualization techniques, we will attempt to better understand the patterns and trends in the data and present them in a visual format.

2.1 PROBLEM STATEMENT

- **Customer Purchasing Behaviour:**
 - Identify top-selling products and buying patterns.
 - Segment customers based on purchase frequency and preferences.
 - Detect seasonal trends and demand fluctuations.
 - Discover frequently purchased item combinations to support product recommendation and promotions.
- **Identify Sales Patterns and Regional Trends:**
 - Analyze sales performance across different regions and store locations.
 - Identify peak sales periods.
 - Measure the impact of promotions & pricing strategies.
 - Forecast future sales trends to support planning and growth strategies.
- **Inventory Optimization:**
 - Improve stock management by identifying fast-moving and slow-moving products.
 - Reduce stock shortages and overstocking.
 - Align inventory levels with customer demand trends.
- **Customer Satisfaction & Business Growth:**
 - Enhance customer loyalty through personalized promotions.
 - Optimize product assortment based on demand analysis.
 - Identify potential locations for business expansion.

2.2 OBJECTIVE

- **Identify key factors driving sales:** Analyze data to identify factors that drive sales.
- **Optimize inventory management:** Develop strategies to optimize inventory management by analyzing data.
- **Improve customer satisfaction:** Develop strategies to improve customer satisfaction by analyzing data.
- **Predict sales:** Use machine learning models to predict sales.
- **Discover frequent purchase patterns:** Use association rule mining to identify product combinations often bought together.
- **Inform business decisions:** Use data analysis results to inform business decisions.

2.3 SCOPE

- Monthly sales & profit to identify seasonal patterns, peak periods, and fluctuations.
- Identify top 10 cities by sales and compare category-wise performance.
- Determine the top 5 customers based on purchase frequency to understand customer loyalty.
- Analyze monthly sales trends by category to observe trends over time.
- Discover frequent itemsets (products often bought together) for targeted recommendations.
- Develop a predictive model to forecast sales for the next 12 months for strategic planning and inventory management.
- Identify key factors influencing sales to reveal what drives profit and loss.
- Assess the impact of discounts on profit and sales to inform pricing and marketing decisions.

2.4 TECHNOLOGY

- Platform – Windows 11
Version – 23H2



- Tool – Excel
Version – 19H2



- Tool – Python
Version – 3.13



- Tool – Jupyter
Version – 7.3.2



3. DATA COLLECTION

3.1 DATA COLLECTION METHOD

1. Supermart Grocery sales – Retail Analytics Dataset
<https://www.kaggle.com/datasets/mohamedharris/supermart-grocery-sales-retail-analytics-dataset/data>
2. Form of Data
 - Data is in csv form having quantitative as well as qualitative data.
3. Rows & Columns

```
df.shape  
  
(9994, 11)
```

3.2 DATA CLEANING & PRE-PROCESSING

- Here the column that are highlighted Yellow is not needed so we have deleted them.

Supermart Grocery Sales - Retail Analytics Dataset - Excel

Order ID	Customer Name	Category	Sub Category	City	Order Date	Region	Sales	Discount	Profit	State
OD1	Harish	Oil & Masala	Masalas	Vellore	08-11-2017	North	1254	0.12	401.28	Tamil Nadu
OD2	Sudha	Beverages	Health Drinks	Krishnagiri	08-11-2017	South	749	0.18	149.8	Tamil Nadu
OD3	Hussain	Food Grains	Atta & Flour	Perambalur	12-06-2017	West	2360	0.21	165.2	Tamil Nadu
OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	11-10-2016	South	896	0.25	89.6	Tamil Nadu
OD5	Ridhesh	Food Grains	Organic Staples	Ooty	11-10-2016	South	2355	0.26	918.45	Tamil Nadu
OD6	Adavan	Food Grains	Organic Staples	Dharmapuri	09-06-2015	West	2305	0.26	322.7	Tamil Nadu
OD7	Jonas	Fruits & Veggies	Fresh Vegetables	Trichy	09-06-2015	West	826	0.33	346.92	Tamil Nadu
OD8	Hafiz	Fruits & Veggies	Fresh Fruits	Ramanadhapuram	09-06-2015	West	1847	0.32	147.76	Tamil Nadu
OD9	Hafiz	Bakery	Biscuits	Tirunelveli	09-06-2015	West	791	0.23	181.93	Tamil Nadu
OD10	Krithika	Bakery	Cakes	Chennai	09-06-2015	West	1795	0.27	484.65	Tamil Nadu
OD11	Ganesh	Snacks	Chocolates	Karur	09-06-2015	West	1903	0.13	437.69	Tamil Nadu
OD12	Yadav	Eggs, Meat & Fish	Eggs	Namakkal	09-06-2015	West	701	0.1	308.44	Tamil Nadu
OD13	Sharon	Snacks	Cookies	Dindigul	15-04-2018	South	1659	0.19	315.21	Tamil Nadu
OD14	Peer	Fruits & Veggies	Fresh Vegetables	Kanyakumari	05-12-2017	West	1277	0.25	63.85	Tamil Nadu
OD15	Sundar	Eggs, Meat & Fish	Chicken	Kanyakumari	22-11-2016	Central	831	0.22	207.75	Tamil Nadu
OD16	Ramesh	Oil & Masala	Edible Oil & Ghee	Krishnagiri	22-11-2016	Central	1440	0.11	100.8	Tamil Nadu
OD17	Alan	Bakery	Cakes	Dharmapuri	11-11-2015	Central	1678	0.28	318.82	Tamil Nadu
OD18	Arutra	Beverages	Health Drinks	Bodi	13-05-2015	West	1617	0.19	113.19	Tamil Nadu
OD19	Haseena	Eggs, Meat & Fish	Mutton	Tenkasi	27-08-2015	West	1757	0.35	386.54	Tamil Nadu

- We have added one Column Order_Month base on Order_Date that is important for our data.

SuperMart groceries sales - Excel

Order ID	Customer Name	Category	Sub Category	City	Order Date	Order_Month	Region	Sales	Discount	Profit
OD1	Harish	Oil & Masala	Masalas	Vellore	08-11-2017	11-2017	North	1254	0.12	401.28
OD2	Sudha	Beverages	Health Drinks	Krishnagiri	08-11-2017	11-2017	South	749	0.18	149.8
OD3	Hussain	Food Grains	Atta & Flour	Perambalur	12-06-2017	06-2017	West	2360	0.21	165.2
OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	11-10-2016	10-2016	South	896	0.25	89.6
OD5	Ridhesh	Food Grains	Organic Staples	Ooty	11-10-2016	10-2016	South	2355	0.26	918.45
OD6	Adavan	Food Grains	Organic Staples	Dharmapuri	09-06-2015	06-2015	West	2305	0.26	322.7
OD7	Jonas	Fruits & Veggies	Fresh Vegetables	Trichy	09-06-2015	06-2015	West	826	0.33	346.92
OD8	Hafiz	Fruits & Veggies	Fresh Fruits	Ramanadhapuram	09-06-2015	06-2015	West	1847	0.32	147.76
OD9	Hafiz	Bakery	Biscuits	Tirunelveli	09-06-2015	06-2015	West	791	0.23	181.93
OD10	Krithika	Bakery	Cakes	Chennai	09-06-2015	06-2015	West	1795	0.27	484.65
OD11	Ganesh	Snacks	Chocolates	Karur	09-06-2015	06-2015	West	1903	0.13	437.69
OD12	Yadav	Eggs, Meat & Fish	Eggs	Namakkal	09-06-2015	06-2015	West	701	0.1	308.44
OD13	Sharon	Snacks	Cookies	Dindigul	15-04-2018	04-2018	South	1659	0.19	315.21
OD14	Peer	Fruits & Veggies	Fresh Vegetables	Kanyakumari	05-12-2017	12-2017	West	1277	0.25	63.85
OD15	Sundar	Eggs, Meat & Fish	Chicken	Kanyakumari	22-11-2016	11-2016	Central	831	0.22	207.75
OD16	Ramesh	Oil & Masala	Edible Oil & Ghee	Krishnagiri	22-11-2016	11-2016	Central	1440	0.11	100.8
OD17	Alan	Bakery	Cakes	Dharmapuri	11-11-2015	11-2015	Central	1678	0.28	318.82
OD18	Arutra	Beverages	Health Drinks	Bodi	13-05-2015	05-2015	West	1617	0.19	113.19
OD19	Haseena	Eggs, Meat & Fish	Mutton	Tenkasi	27-08-2015	08-2015	West	1757	0.35	386.54
OD20	Haseena	Beverages	Soft Drinks	Kanyakumari	27-08-2015	08-2015	West	693	0.38	159.16

SuperMart grocery sales - Excel

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Order_ID	Customer Name	Category	Sub_Category	City	Order_Date	Order_Month	Region	Sales	Discount	Profit					
2	OD1	Harish	Oil & Masala	Masalas	Vellore	08-11-2017	11-2017	North	1254	0.12	401.28					
3	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	08-11-2017	11-2017	South	749	0.18	149.8					
4	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	12-06-2017	06-2017	West	2360	0.21	165.2					
5	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	11-10-2016	10-2016	South	896	0.25	89.6					
6	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	11-10-2016	10-2016	South	2355	0.26	918.45					
7	OD6	Adavan	Food Grains	Organic Staples	Dharmapuri	09-06-2015	06-2015	West	2305	0.26	322.7					
8	OD7	Jonas	Fruits & Veggies	Fresh Vegetables	Trichy	09-06-2015	06-2015	West	826	0.33	346.92					
9	OD8	Hafiz	Fruits & Veggies	Fresh Fruits	Ramanadhapuram	09-06-2015	06-2015	West	1847	0.32	147.76					
10	OD9	Hafiz	Bakery	Biscuits	Tirunelveli	09-06-2015	06-2015	West	791	0.23	181.93					
11	OD10	Krithika	Bakery	Cakes	Chennai	09-06-2015	06-2015	West	1795	0.27	484.65					
12	OD11	Ganesh	Snacks	Chocolates	Karur	09-06-2015	06-2015	West	1903	0.13	437.69					
13	OD12	Yadav	Eggs, Meat & Fish	Eggs	Namakkal	09-06-2015	06-2015	West	701	0.1	308.44					
14	OD13	Sharon	Snacks	Cookies	Dindigul	15-04-2018	04-2018	South	1659	0.19	315.21					
15	OD14	Peer	Fruits & Veggies	Fresh Vegetables	Kanyakumari	05-12-2017	12-2017	West	1277	0.25	63.85					
16	OD15	Sundar	Eggs, Meat & Fish	Chicken	Kanyakumari	22-11-2016	11-2016	Central	831	0.22	207.75					
17	OD16	Ramesh	Oil & Masala	Edible Oil & Ghee	Krishnagiri	22-11-2016	11-2016	Central	1440	0.11	100.8					
18	OD17	Alan	Bakery	Cakes	Dharmapuri	11-11-2015	11-2015	Central	1678	0.28	318.82					
19	OD18	Arutra	Beverages	Health Drinks	Bodi	13-05-2015	05-2015	West	1617	0.19	113.19					
20	OD19	Haseena	Eggs, Meat & Fish	Mutton	Tenkasi	27-08-2015	08-2015	West	1757	0.35	386.54					
21	OD20	Urmila	Beverages	Soft Drinks	Kanyakumari	27-08-2015	08-2015	West	683	0.26	158.18					

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- We used python library pandas. We import csv file into (df) dataframe. Using read_csv method.

```
import pandas as pd
df = pd.read_csv(r"C:\Users\Public\Downloads\SuperMart grocery sales.csv")
df
```

	Order_ID	Customer Name	Category	Sub_Category	City	Order_Date	Order_Month	Region	Sales	Discount	Profit
0	OD1	Harish	Oil & Masala	Masalas	Vellore	08-11-2017	11-2017	North	1254	0.12	401.28
1	OD2	Sudha	Beverages	Health Drinks	Krishnagiri	08-11-2017	11-2017	South	749	0.18	149.80
2	OD3	Hussain	Food Grains	Atta & Flour	Perambalur	12-06-2017	06-2017	West	2360	0.21	165.20
3	OD4	Jackson	Fruits & Veggies	Fresh Vegetables	Dharmapuri	11-10-2016	10-2016	South	896	0.25	89.60
4	OD5	Ridhesh	Food Grains	Organic Staples	Ooty	11-10-2016	10-2016	South	2355	0.26	918.45
...	...	...	...	...	...	...	...	...	...	...	...
9989	OD9990	Sudeep	Eggs, Meat & Fish	Eggs	Madurai	24-12-2015	12-2015	West	945	0.16	359.10
9990	OD9991	Alan	Bakery	Biscuits	Kanyakumari	12-07-2015	07-2015	West	1195	0.26	71.70
9991	OD9992	Ravi	Food Grains	Rice	Bodi	06-06-2017	06-2017	West	1567	0.16	501.44
9992	OD9993	Peer	Oil & Masala	Spices	Pudukottai	16-10-2018	10-2018	West	1659	0.15	597.24
9993	OD9994	Ganesh	Food Grains	Atta & Flour	Tirunelveli	17-04-2018	04-2018	West	1034	0.28	165.44

9994 rows x 11 columns


```
print(df.dtypes)
```

Order_ID	object
Customer Name	object
Category	object
Sub_Category	object
City	object
Order_Date	object
Order_Month	object
Region	object
Sales	int64
Discount	float64
Profit	float64
dtype:	object



```
df["Order_Date"] = pd.to_datetime(df["Order_Date"], format="%d-%m-%Y")
df["Order_Month"] = df["Order_Date"].dt.to_period("M").dt.to_timestamp()
print(df.dtypes)
```

Order_ID	object
Customer Name	object
Category	object
Sub_Category	object
City	object
Order_Date	datetime64[ns]
Order_Month	datetime64[ns]
Region	object
Sales	int64
Discount	float64
Profit	float64
dtype:	object

- Here, we checked data is null or not using isnill () function.

```
print(df.isnull().sum())
```

Order_ID	0
Customer Name	0
Category	0
Sub_Category	0
City	0
Order_Date	0
Order_Month	0
Region	0
Sales	0
Discount	0
Profit	0
dtype:	int64

4. EXPLORATORY DATANALYSIS

- Through a comprehensive Exploratory Data Analysis of the supermarket grocery sales dataset, we uncovered valuable patterns, seasonality trends, and areas of profitability. This analysis provided actionable insights into customer habits and preferences, enabling the development of targeted strategies to drive business growth, enhance customer loyalty, and improve retention. By leveraging data-driven decision-making, we can recommend relevant products, boost sales, and create a more personalized shopping experience.

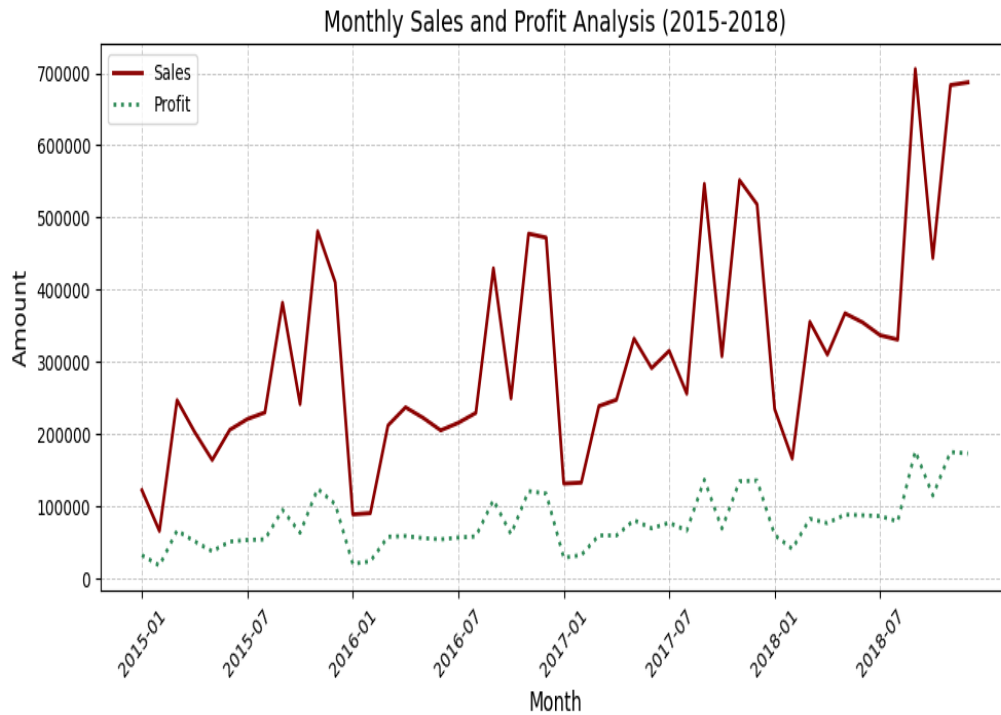
4.1 SUMMARIZING STATISTICS

- Here, we define Statically information of quantitative data. Like min, max, mean count, standard deviation.

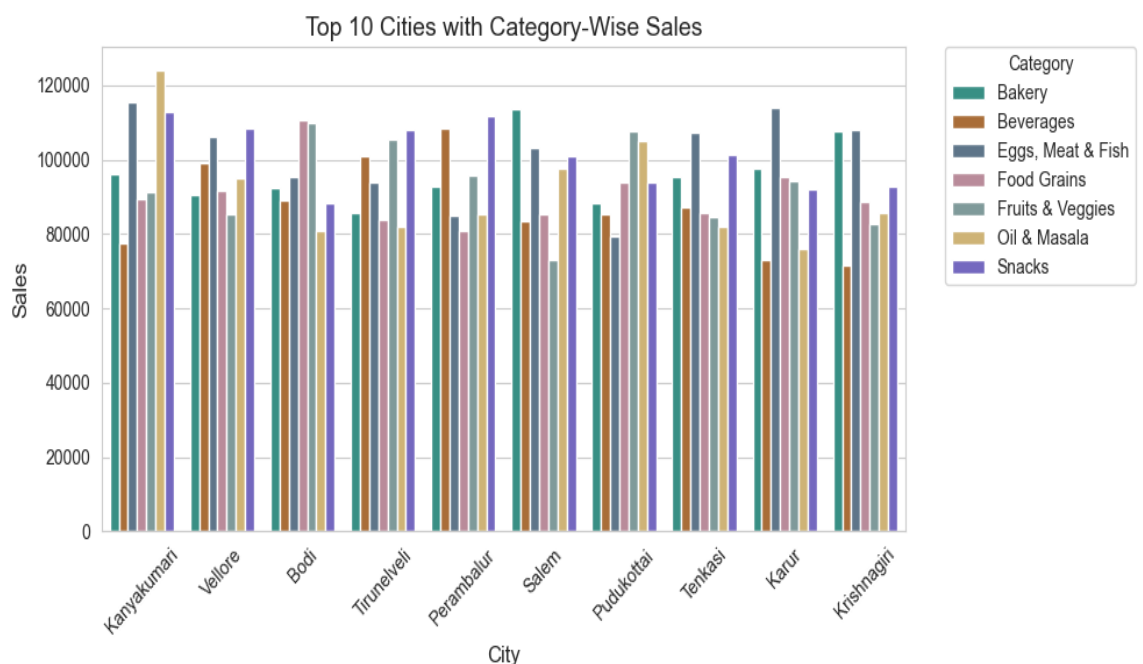
	Sales	Discount	Profit
count	9994.000000	9994.000000	9994.000000
mean	1496.596158	0.226817	374.937082
std	577.559036	0.074636	239.932881
min	500.000000	0.100000	25.250000
25%	1000.000000	0.160000	180.022500
50%	1498.000000	0.230000	320.780000
75%	1994.750000	0.290000	525.627500
max	2500.000000	0.350000	1120.950000

4.2 DATA VISUALIZATION

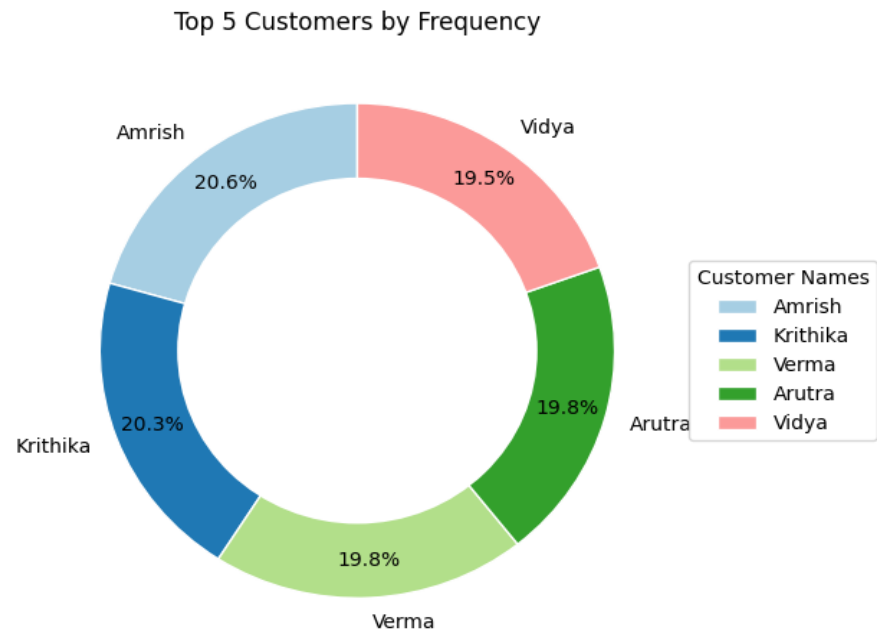
- **Sales & Profit by Month:** Monthly trends in sales and profit were analyzed. This helped identify seasonal patterns, peak sales periods, and fluctuations. This visualization supports forecasting and effective business planning.



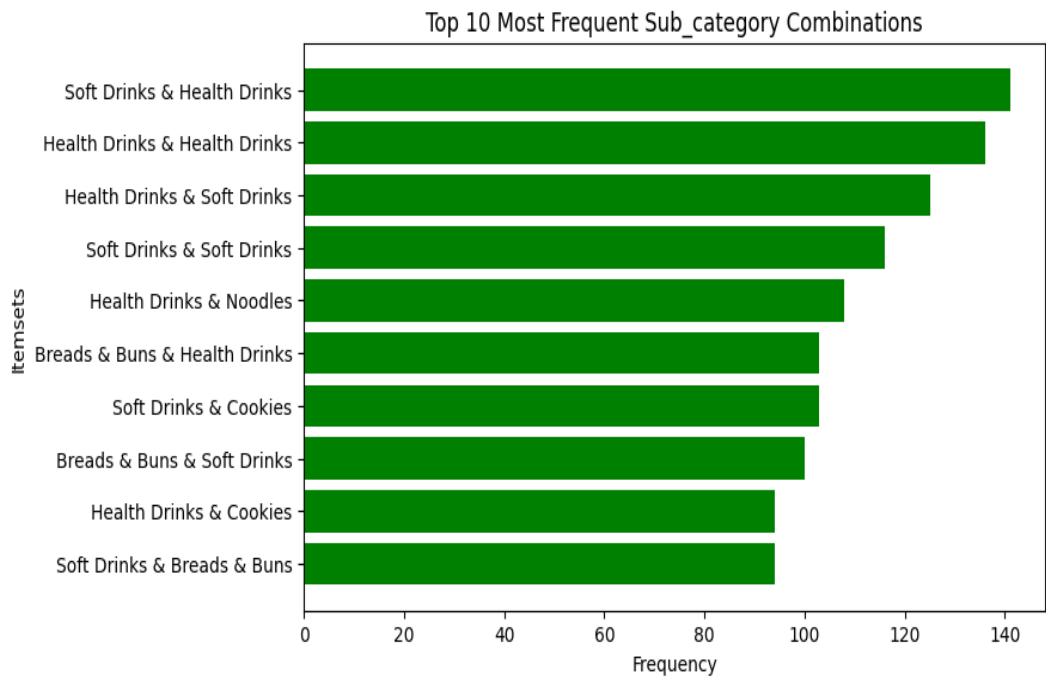
- **Top 10 Performing Cities with Category-Wise Sales:** Identified cities contributing the most to sales. Optimize Underperforming areas & compare city-wise Category sales.



- **Top 5 Customers by Frequency Analysis:** Top 5 customers were identified based on purchase frequency. This analysis helps in understanding customer loyalty and recognizing repeat buying behavior.

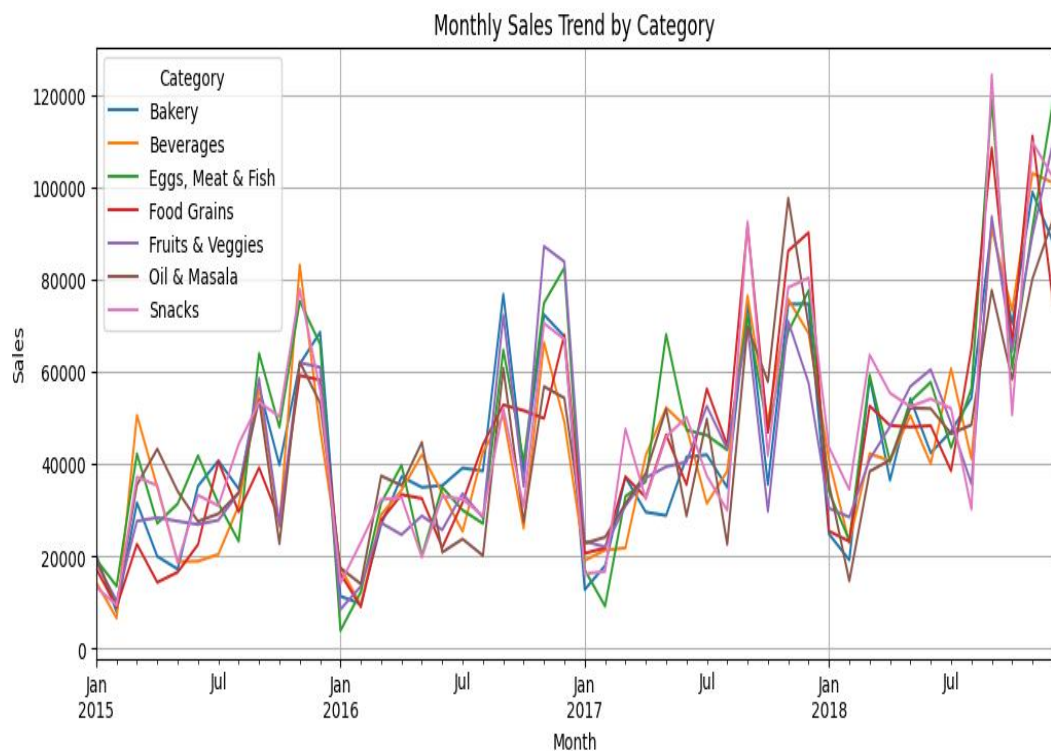


- **Frequent itemsets:** We discovered item combinations frequently purchased together by customers. This analysis helps recommend relevant products, improve sales, and enhance the shopping experience.



4.3 Predictive Modeling

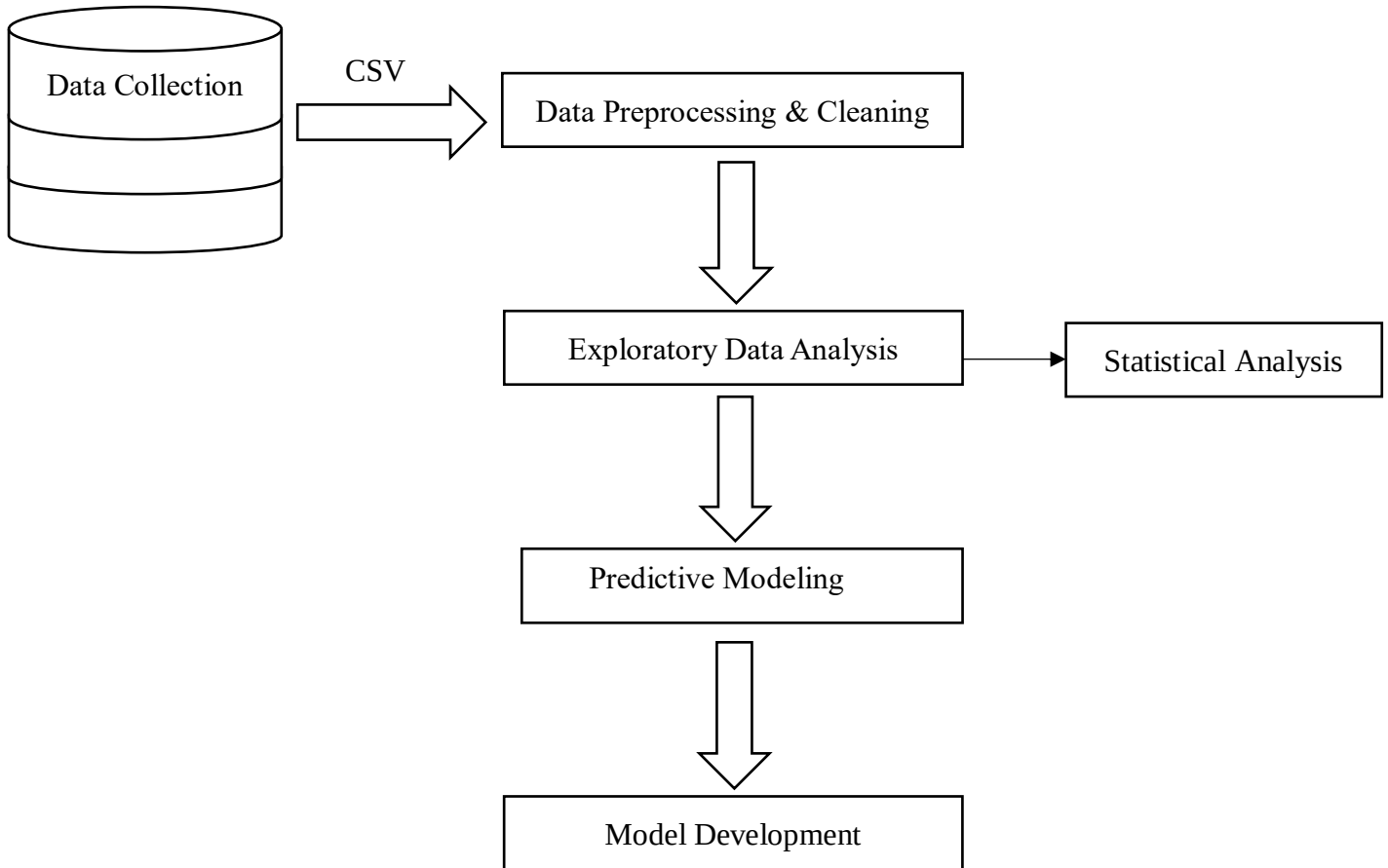
- We implemented Predictive Modeling techniques using historical sales data to forecast future demand and uncover emerging trends. This helped improve inventory planning, optimize stock levels, and align supply with projected demand, empowering the business to make forward-looking, data-driven decisions for maximizing sales opportunities.
- **Monthly Sales Trends by Category:** Analyzed Sales data for each category to observe sales trends over time



- **Sales Forecasting for the Next 12 Months:** Developed a predictive model to forecast sales for the next 12 months. This helps in strategic planning, inventory management, and identifying potential growth opportunities.



5. METHODOLOGY



Data Collection:

- We have collected data from Keggles.

Data Pre-processing & Cleaning:

- We dropped unnecessary column, such as State (Tamil Nadu).
- We added new columns Order_Month based on Order_Date.
- We converted Order_Date and Order_Month columns to datetime format.
- We checked for non-values in the data.

Exploratory Data Analysis (EDA):

- In pandas' library we have used describe method to display basic statistical information.
- **Sales & Profit by Month:** Monthly trends in sales and profit were analyzed using a line chart. This helped identify seasonal patterns, peak sales periods, and fluctuations. This visualization supports forecasting and effective business planning.
- **Top 10 Performing Cities with Category-Wise Sales:** A scatter bar chart identified cities contributing the most to sales. Optimize underperforming areas compare city-wise category sales.
- **Top 5 Customers by Frequency Analysis:** A donut chart was used to identify the top 5 customers based on purchase frequency. This analysis helps in understanding customer loyalty and recognizing repeat buying behavior.
- **Monthly Sales Trends by Category:** Line chart was employed to analyze sales data for each category, observing sales trends over time.
- **Key Factors Influencing Sales:** Linear regression analysis revealed the impact of various factors on sales, enabling the identification of key elements that drive profit or loss.

Used Algorithm:

- **Frequent Itemsets:** Association rule mining was utilized to discover item combinations frequently purchased together by customers. This analysis enables the recommendation of relevant products, improves sales, and enhances the shopping experience.
- **Sales Forecasting for The Next 12 Months:** An SARIMA model was developed to forecast sales for the next 12 months. This facilitates strategic planning, inventory management, and identification of potential growth opportunities.
- **Impact of Discount on Profit and Sales:** K-means clustering analysis demonstrated that discounts significantly impact profit and sales, informing better pricing and marketing decisions.

6. FEATURE ENGINEERING

6.1 EXPLANATION OF FEATURE SELECTION AND EXTRACTION

- By analyzing fields like Order_Month, Sales, Profit, Category, City, Customer Name, and Sub-Category, we uncovered key patterns, trends in customer behavior and sales performance. These insights helped identify profitable areas, improve customer retention, and support data-driven strategies for business growth. Additionally, we discovered co-purchased item combinations, supporting data-driven decisions on inventory, bundling, and recommendations to boost sales and customer satisfaction.
- Using fields such as Category, Sales, and Order_Date(M) This helped improve inventory planning, optimize stock levels, and align supply with projected demand, empowering the business to make forward-looking, data-driven decisions for maximizing sales opportunities.
- With features like Discount, Profit, Category, Region, and Sales, we identified the key factors influencing sales and profitability. The analysis revealed the impact of discounts and guided better pricing and marketing decisions, ultimately supporting smarter strategies and increased business efficiency.

6.2 DERIVED FEATURE

- With features like Discount, Profit, Category, Region, and Sales, we identified the key factors influencing sales and profitability. The analysis revealed the impact of discounts and guided better pricing and marketing decisions, ultimately supporting smarter strategies and increased business efficiency.
- **Order_Month:** We created the "Order Month" column based on the "Order Date" column, so that I can analyze the sales made in different months.

- **Formula:**

=TEXT(F2,"MM-YYYY")

- **Improving business sales strategies:** This analysis helps business sales strategies.
- **Informing future planning:** Forecasting sales for the next 12 months aid in planning business's future.
- **Understanding financial performance:** Analyzing monthly sales and profit helps understand business's financial situation.
-

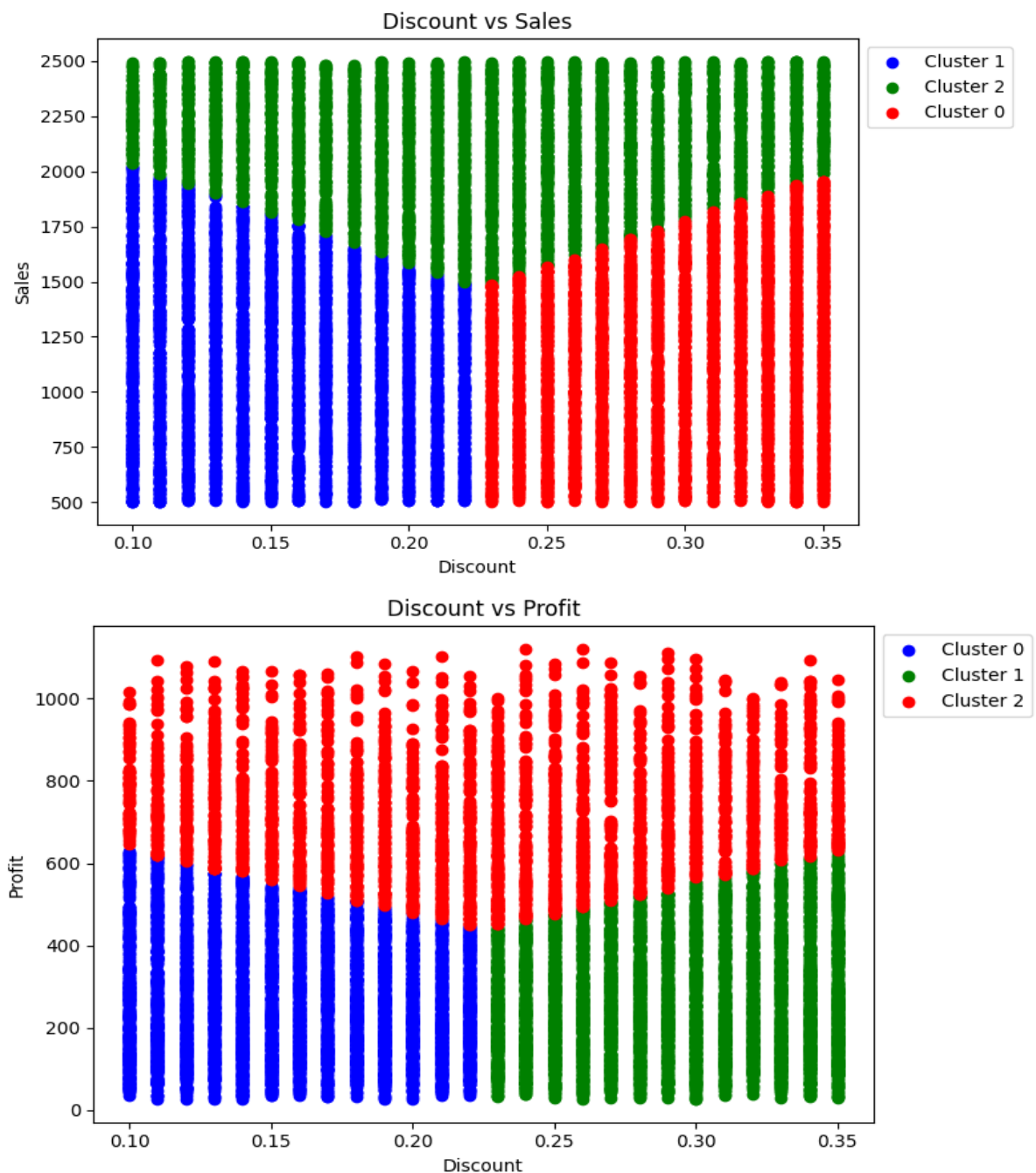
6.3 FEATURE SCALING

We used StandardScaler to bring the Discount vs Sales, Discount vs Profit features to the same scale. This is important because KMeans clustering is a distance-based algorithm. If the features have different scales, one feature might have more influence than the others, which could affect the clustering results. By scaling the features, I ensured that all of them contribute equally to the clustering process, so no single feature dominates. This approach helps in achieving more balanced and accurate clustering results.

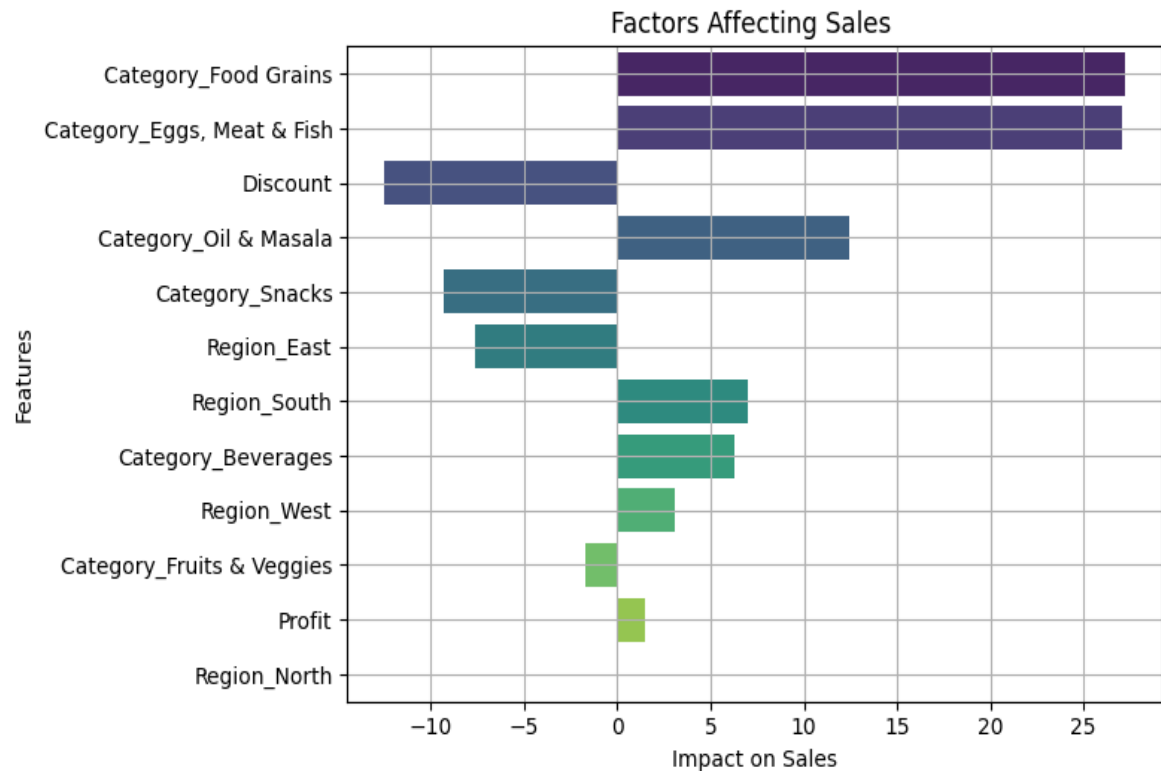
7. Model Development

During Model Development, we identified and understand their influence on sales performance. The models revealed the critical factors driving profit and loss, while also evaluating how pricing strategies and discounts impact revenue. These insights helped refine marketing and pricing decisions, ultimately enhancing overall business efficiency and profitability.

- **Impact of Discount on Sales and Profit:** The analysis shows that discounts significantly impact profit and sales, supporting better pricing and marketing decisions.



- **Key Factors Influencing Sales:** It shows the impact of various types of factors on sales. The purpose is to identify key elements that lead to profit or loss. This analysis helps in making better business decisions.



8. Evaluation

- The SARIMAX (1,1,1) (1,1,1,12) model effectively captures both trend and seasonality in the sales data. All coefficients are statistically significant, indicating model stability. The Ljung-Box test p-value of 0.15 suggests no significant autocorrelation in residuals, while the Jarque-Bera test p-value of 0.46 indicates normality in residuals. Additionally, the heteroskedasticity test p-value of 0.75 shows no significant heteroskedasticity, confirming the model's robustness.

SARIMAX Results						
=====						
Dep. Variable:	Sales		No. Observations:		48	
Model:	SARIMAX(1, 1, 1)x(1, 1, 1, 12)		Log Likelihood		-422.541	
Date:	Mon, 14 Apr 2025		AIC		855.082	
Time:	22:47:46		BIC		862.859	
Sample:	01-31-2015		HQIC		857.767	
	- 12-31-2018					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

ar.L1	-0.6653	0.205	-3.246	0.001	-1.067	-0.264
ma.L1	0.4170	0.229	1.818	0.069	-0.033	0.867
ar.S.L12	-0.8827	0.119	-7.393	0.000	-1.117	-0.649
ma.S.L12	0.9998	0.322	3.108	0.002	0.369	1.630
sigma2	1.604e+09	2.01e-10	7.96e+18	0.000	1.6e+09	1.6e+09
=====						
Ljung-Box (L1) (Q):	2.11	Jarque-Bera (JB):		1.54		
Prob(Q):	0.15	Prob(JB):		0.46		
Heteroskedasticity (H):	1.21	Skew:		0.39		
Prob(H) (two-sided):	0.75	Kurtosis:		2.34		
=====						

- A Silhouette Score of 0.59 is considered good. It means that your KMeans clustering model has effectively captured the pattern between Discount and Sales.

Silhouette Score for KMeans Clustering: 0.59

- A Silhouette Score of 0.58 is considered good. It means that your KMeans clustering model has effectively captured the pattern between Discount and Profit.

Silhouette Score for KMeans Clustering: 0.58

9. CONCLUSION

9.1 SUMMARY OF ACHIEVEMENTS

- Uncovered valuable business insights that supported strategic decision-making, long-term growth.
- Revealed critical patterns in customer behavior and sales performance to improve efficiency and profitability.
- Enhanced customer engagement through better understanding of customer needs. Supported sustainable business success through actionable insights.
- Enabled forecasting of trends to anticipate market shifts and optimize planning.
- Identified key business drivers that empowered targeted strategies.
- Discovered patterns that laid the groundwork for improved personalization.
- Strengthened the foundation for data-driven growth.

9.2 IMPLICATION OF FINDING

- Discovered key sales and customer behavior patterns that help understand market trends and shifting demands.
- Identified areas of improvement in product performance and customer engagement for business growth.
- Gained insights into customer needs and preferences, supporting more personalized services.
- Uncovered buying patterns and frequent item associations that can guide smarter product placement.
- Enabled better decision-making by revealing what drives profit and where sales can be optimized.
- Strengthened data-driven strategies to stay aligned with market needs and improve business efficiency.

9.3 LIMITATION

- **Limited Geographical Scope:** The analysis is based only on sales data from Tamil Nadu, so results cannot be generalized to other regions.
- **No Real-Time Data:** The data is not live. It is historical and cannot respond to sudden changes in the market.
- **Missing Customer Demographics:** Details like age, gender, and income of customers are not available, so deep customer behavior analysis is not possible.
- **No Advanced Neural Network Models:** Complex models like RNN or LSTM were not used due to limited data and processing power.

9.4 FUTURE WORK

- **Cross-Selling Recommendation System:** Based on Customer ID and past purchase patterns, a recommendation model can be developed to suggest frequently bought-together items. This can help increase average order value and enhance customer engagement.
- **Customer Churn Prediction:** Using Order Date and purchase frequency, a churn model can be built to identify customers who are likely to stop buying in the near future. It can help initiate timely re-engagement campaigns and improve retention.
- **Customer Networking or Similarity Clustering:** By analyzing sales patterns, customers can be grouped into clusters based on similar behavior or preferences. These clusters can be used to personalize marketing, offer group-based deals, or build referral networks.

10. REFERENCES

Kaggle Link:

<https://www.kaggle.com/datasets/mohamedharris/supermart-grocery-sales-retail-analytics-dataset/data>

GitHub Link:

https://github.com/Varshachoudhary05/Colorado_motor_Vehicle_sales